

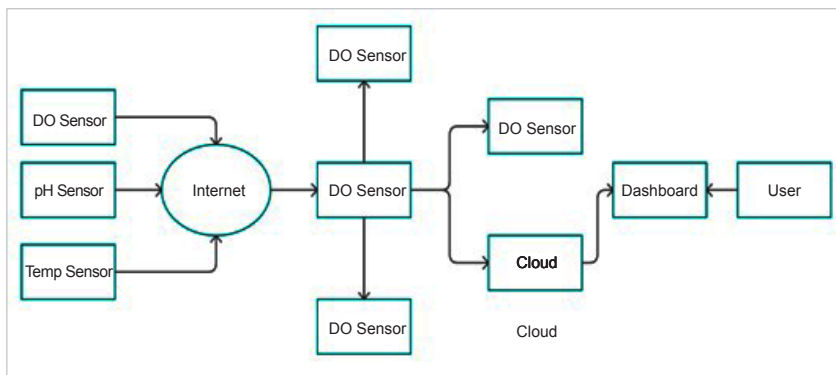


Figure 1: Block diagram of iAqurium

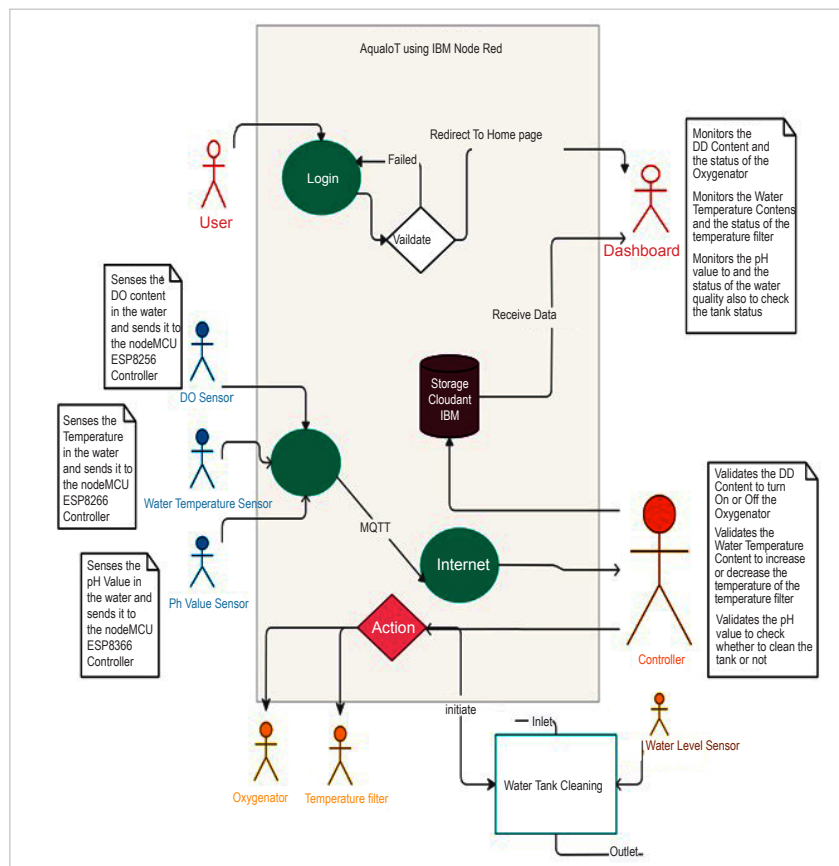


Figure 2: Use case diagram of iAqurium

used to measure the water temperature and data is used to control it. The normal and optimal water temperature for most fish lies between 21°C and 26°C. This automation system is designed with sensor modules (input data), sensor data action (output), cleaning of the tank, along with cloud integration and a Web interface. The

entire system is hosted in the cloud and allows both Web and mobile interfaces, so the user can monitor it from any part of the world. It is simulated using the Node-RED environment with the IBM Cloudant database. We will call this system iAqurium.

iAqurium system design

Figure 1 represents the block diagram

of the iAqurium system with all its functionalities. The use case diagram is depicted in Figure 2.

Steps to design the iAqurium

- Install the latest version of Node.js from <https://nodejs.org/en/download/>.
- Install Node-RED using the command `npm install -g node-red`.
- From the browser, get access to the Node-RED flow using the IP `http://127.0.0.1:1880`.
- From the Node palette of Node-RED, insert the appropriate nodes, configure their properties and wire them.
- Click the *Deploy* button and confirm deployment.
- Visualise the output through the *Debug* monitor window and dashboard UI.

Node flow design of iAqurium

1. Insert *inject* node to trigger the event.
2. Insert *function* node and write the appropriate JavaScript code for automation.
3. Insert *join* node and *MQTT* node to transmit alert messages through HiveMQ.
4. Insert *cloudant out* node to push data onto the cloud.
5. Insert *template* node to create a UI design.
6. Insert *debug* node for printing msg.payload.

iAqurium working modules

Sensor data: This module involves data collection from all the sensors in the system like pH sensor, DO sensor, and water temperature sensor. The sensor values are generated in Node-RED seamlessly at regular intervals of 3 seconds, as shown in Figure 3. The data is then sent to the controller, through the *MQTT out* node using the HiveMQ broker. The DO content is measured in ppm units and the water temperature in Celsius.